Chronos Roadmap

v17 · 2026-10-21 · replaces the v16 roadmap · theme: Bureau CRT terminal (phosphor line art, ASCII, dither)

How to read this

Size: S = under one session, M = one or two, L = three or more or needs compositor work. Layer: shell = Quickshell QML, niri = compositor fork, nix = MyNixOS, art = assets. (T) = needs the Surface to test.

Vocab in use: Timeline = workspace, Dossier = window, Archive = Overview, Index = spotlight, Memos = notifications, Controls = quick settings, Dock, Clearance = lock, Office = desktop mode, Field = tablet mode. Proposed new words: Ledger = clipboard history, Capture = screenshot, Transmit = screen recording, Telemetry = system monitor, Cache = scratchpad, Redaction = error state.

0. Done and tested

- Stack: Quickshell + QML shell and niri fork in one repo (ChronosWM), deployed through the MyNixOS flake input `chronos`.
- Theme: Terminal phosphor default, one hue per Timeline, Office and Dreamcore alternates (partial), Departure Mono, scanlines.
- Gestures: four edge swipes with finger-following panels, 3f scroll and Timeline switch, long bottom swipe opens the Archive.
- Panels: Controls (wifi, bluetooth, light, volume, battery, silence, mode, theme), Index (app search, calculator), Memos, Dock (open windows), Archive strip.
- ASCII components: Frame, DataRow, BarRow, AsciiIcon.
- OSK: editable Keychron-style layouts, es nodeadkeys default, terminal print-in animation, types into Index or through wtype.
- Mode switch: Office and Field, Field by default on the Surface, toggle in Controls or `chronos-shell ipc mode toggle`.
- Cursor hides on its own. Mock IPC and gesture-core tests run on the desktop.
- Still untested: 4f quick-switch and the 4f-up Archive event.

Dependency map

```
[Palm/dead zones] --> [Pen support] --> [Markup]
[Rotation] --> [Auto Field/Office] --> [OSK v2]
[Text-input hook] --> [OSK auto-show] --> [Clearance PIN entry]
[Clearance] --> [Idle chain] --> [Power menu]
[Timeline model] --> [Named Timelines] --> [Archive v2] --> [Session restore]
[Theme tokens] --> [CRT shaders] --> [Dreamcore content]
[Theme tokens] --> [GTK/Qt/terminal theming]
```

1. Daily driver (the Surface can replace GNOME)

Session and security

ID	Item	Layer	Size
1.01	Clearance lock screen: session-lock protocol, dossier-style, OSK for password, tinted by Timeline hue (T)	shell	M
1.02	Idle chain: dim, blank, Clearance, suspend; configurable delays	shell + nix	M
1.03	Power menu as a confirm Dossier (power off, reboot, suspend, hibernate); power button short and long press	shell	S
1.04	Themed greeter with OSK (SDDM theme or greetd + Quickshell)	nix + shell	L
1.05	Themed polkit authentication prompt	shell	M
1.06	Keyring / secret service unlocked at login	nix	S
1.07	Safe-mode fallback: if the QML fails to load, a minimal recovery panel with restart and log	shell	S

Input and Field mode

ID	Item	Layer	Size
1.10	Palm rejection and dead zones (edge band vs palm, suppress touch while the pen is near) (T)	niri	L
1.11	Rotation from the accelerometer, rotation lock in Controls, touch and OSK follow (T)	niri + shell	M

ID	Item	Layer	Size
1.12	OSK auto-show on text-field focus in any app (text-input hook in the fork, event to the shell)	niri + shell	L
1.13	Type Cover attach/detach switches Office and Field automatically (manual toggle stays) (T)	niri + shell	M
1.14	Gesture thresholds as live settings: edge band, claim window, commit progress, velocity	niri + shell	S
1.15	Test and fix the 4f quick-switch and the 4f-up Archive event (T)	niri	S

Connectivity and media

ID	Item	Layer	Size
1.20	Wifi picker: scan, join with OSK password, forget	shell	M
1.21	Bluetooth devices: list, pair, connect, disconnect	shell	M
1.22	Audio output and input switching, mic mute, per-app volume	shell	M
1.23	Volume and brightness OSD as an ASCII bar that auto-hides	shell	S
1.24	Media keys and MPRIS controls (play, pause, next) in Controls	shell	S

Daily tools

ID	Item	Layer	Size
1.30	Capture: screenshot full, region, Dossier; saves and copies	shell	M
1.31	Ledger: clipboard history, paste from Index	shell + nix	M
1.32	Memos history: log, actions, urgency, DND schedule	shell	M
1.33	Status line: thin optional strip with clock, battery, SIGNAL, Timeline	shell	M
1.34	System tray (StatusNotifier) for apps that need it	shell	S

ID	Item	Layer	Size
1.35	XWayland for legacy apps (xwayland-satellite)	nix + niri	S
1.36	XDG portals: file chooser, screencast, screenshot, dark-mode setting	nix	M
1.37	Window rules per app-id: Timeline, float, size	niri + nix	M
1.38	Dock v2: pinned apps, running indicators, launch	shell	M
1.39	Autostart list and session restore (reopen Dossiers on their Timelines)	shell + nix	M
1.40	Retire dynisland (remove package and unit)	nix	S

2. Comfortable (a WM you enjoy using)

Window management

ID	Item	Layer	Size
2.01	Floating Dossiers and pin (always on top)	niri	M
2.02	Cache: drop-down terminal / scratchpad	niri + shell	M
2.03	Tabbed columns	niri	M
2.04	Dossier switcher (Alt-Tab and 4f hold) showing Dossier cards	shell	M
2.05	Urgency hint: Dossier card blinks in Dock and Archive	shell	S
2.06	Named Timelines: create, delete, rename, reorder	niri + shell	M
2.07	Move a Dossier between Timelines by dragging in the Archive or by keybind	niri + shell	M
2.08	Persistent per-Timeline hue, wallpaper and name	shell	S
2.09	Multi-monitor: hotplug, per-output Timelines, docking profile (USB-C)	niri + nix	L
2.10	Fractional scale per output, scale picker in Controls	niri + shell	M
2.11	Picture-in-picture Dossier	niri	M

ID	Item	Layer	Size
2.12	Office mode: full keybinding set, focus-follows-mouse option, pointer hot edges for the panels	niri + shell	M
2.13	Keybinding cheat sheet in Index (type `?`)	shell	S

Shell panels

ID	Item	Layer	Size
2.20	Index v2: file search, window search, `>` commands, unit and currency calc, glyph picker, settings search, recents	shell	L
2.21	Archive v2: dithered Dossier thumbnails, Timeline rename, drag Dossiers	shell	L
2.22	Controls v2: VPN, airplane, night light, power profile, keyboard layout (es/en), auto-brightness, battery detail, record toggle	shell	L
2.23	Telemetry: CPU, RAM, temperature, network as ASCII waveforms (style of reference 03)	shell	M
2.24	Process list with kill	shell	M
2.25	Agenda dossier (calendar, ICS or CalDAV)	shell	M
2.26	Media panel with dithered cover art	shell	M
2.27	Transmit: screen recording toggle and indicator	shell	M
2.28	Settings Dossier: theme tokens, gesture sensitivity, OSK, mode, hues; applies live	shell	L
2.29	Config hot reload and error banner	shell	S
2.30	Layout indicator and Caps/Num lock chrome	shell	S

Pen and OSK v2

ID	Item	Layer	Size
2.40	Pen events: pressure, eraser, hover, barrel button actions (T)	niri	L
2.41	Markup: draw over screenshots with the pen	shell	M

ID	Item	Layer	Size
2.42	OSK: drag-and-drop key moving, layout rename, per-key shifted output editing	shell	M
2.43	OSK: long-press alternates (á é ñ ü), key repeat on backspace and arrows	shell	M
2.44	OSK: emoji/glyph layer, floating undocked mode, one-handed and split layouts	shell	M
2.45	OSK hides while a physical keyboard is active	shell	S

3. Theme polish (CRT, not too heavy)

ID	Item	Layer	Size
3.01	Theme tokens file `~/.config/chronos/theme.json`, edited live from the Settings Dossier	shell	M
3.02	Glow/bloom shader, grain and subtle flicker as toggles; measure GPU and battery cost on the Surface first (T)	shell	M
3.03	Optional curvature, vignette and chromatic aberration, off by default	shell	S
3.04	Dither pipeline: script that turns images into 1-bit Bayer assets; runtime dither for wallpapers and thumbnails	art + shell	M
3.05	ASCII icon pack growth: spinners, signal bars, battery glyph, tabs, folders, barcode, waveforms	shell	M
3.06	Frame variants: corner ticks, labelled panels, double-line for the Archive, status chrome (SIGNAL LOW, CONNECTED)	shell	M
3.07	Terminal-style transitions: stepped print-in for every panel, typewriter headers, Timeline tint sweep	shell	M
3.08	Dossier borders and focus ring in the Timeline hue; shadows off	niri	S

ID	Item	Layer	Size
3.09	Finish Office (beige paper) and Dreamcore (lilac haze, grainy serif display face) themes	shell	M
3.10	More hues beyond the five, calibrated against the references	shell	S
3.11	GTK, Qt and terminal theming generated from the tokens (alacritty/foot palette follows the hue)	nix	M
3.12	Pixel cursor theme and a monochrome tinted icon fallback	art + nix	M
3.13	Boot and shutdown sequence: console splash in terminal style, SESSION CLOSED on exit	nix	M
3.14	Sound design (opt-in): boot beep, key click, memo tick	shell	M
3.15	Accessibility: reduce motion, high contrast, font scale, large-touch mode	shell	M
3.16	Vocab audit: every UI string uses the Vocab table	shell	S

4. Backrooms and dreamcore content

ID	Item	Layer	Size
4.01	Dithered empty-corridor wallpapers, one per Timeline	art	M
4.02	Odd idle messages, rare and rotating, on empty Timelines and the lock screen	shell	S
4.03	Fluorescent flicker on events (Timeline switch, Memos, lock)	shell	S
4.04	Screensaver: drifting phosphor and text like NO ONE IS HERE	shell	M
4.05	Clearance as a classified dossier with a fake case file	shell	M
4.06	Timeline glitch events: rare, toggleable, never during input	shell	S
4.07	Generated Timeline names (LEVEL 0 and similar)	shell	S

ID	Item	Layer	Size
4.08	Redaction error states: failed actions shown as blacked-out lines	shell	S
4.09	Time-of-day hue shift (dusk toward amber)	shell	S
4.10	Hidden commands in Index (easter eggs)	shell	S
4.11	Ambient fluorescent hum (opt-in)	shell	S

5. Platform and quality

ID	Item	Layer	Size
5.01	CI: build the fork, flake check, qmllint, gesture-core tests on every push	nix	M
5.02	Dependabot triage: merge or close the open cargo and actions branches	niri	S
5.03	QML smoke tests driven by the mock IPC	shell	M
5.04	Debug overlay: window size, panel state, last events (the ANCHOxALTO idea, permanent behind a toggle)	shell	S
5.05	Rebase the fork on upstream niri on a schedule; keep patches to gestures and IPC events	niri	M
5.06	Docs: IPC.md, THEME.md, ROADMAP.md, short user manual under `help` in Index	docs	M
5.07	Config backup and export (OSK layouts, theme tokens, Timeline names)	shell	S
5.08	Release tags and changelog; rollback through NixOS generations	nix	S
5.09	Per-host defaults for desktop and laptop (Office, no edge panels needed)	nix	S

Next up, in order

1. 1.15 Test 4f quick-switch and Archive event.
2. 1.40 Retire dynisland, then 1.07 safe-mode fallback.

3. 1.10 Palm rejection and dead zones, with 1.14 live gesture settings.
4. 1.11 Rotation.
5. 1.12 OSK auto-show in other apps, then 1.13 Type Cover auto mode.
6. 1.01 to 1.03 Clearance, idle chain, power menu.
7. 1.20 to 1.22 Wifi, Bluetooth, audio pickers.
8. 1.30 and 1.31 Capture and Ledger.
9. 1.37 and 1.38 Window rules and Dock v2.
10. 3.01 theme tokens, then 3.02 CRT effects with a cost check.

Open decisions

- Should 4f up toggle niri's Overview (current) or open a shell panel?
- Dock: pinned apps, only running windows, or both?
- Greeter: SDDM theme or greetd with a Quickshell greeter?
- Are the proposed words (Ledger, Capture, Transmit, Telemetry, Cache, Redaction) approved?
- GPU budget for CRT effects on the Surface: always on, only plugged in, or off in Field mode?
- Session restore: which Dossiers come back, and should Timelines keep their names after reboot?